Training deep networks

CIFAR-10

LECTURE 11 | GÉRON CH. 11

Applications of Machine Learning

BSc Mathematics of Artificial Intelligence · Third year

Where we are

You can build a network and you own its training loop. So build a deep one — twenty hidden layers on CIFAR-10 — and train it exactly as Lecture 10 taught.

> **IT DOES NOT TRAIN**
>
> The loss barely moves. Nothing raises, nothing is NaN, the harness is correct and the data is fine. A shallower version of the same network trains without complaint.

Depth is not free, and the reason is arithmetic. Today we measure what goes wrong, derive why, and repair it — in that order, because the repair is unreadable until you have seen the measurement.

Today

1. The failure, **instrumented** — per-layer activation and gradient statistics (20 min)
2. **The mathematics** — variance propagation through layers (20 min)
3. **The method** — initialisation, activations, normalisation, clipping, optimisers, schedules, regularisation (35 min)
4. The ablation: which repair actually paid (15 min)

FIRST, LOOK

A deep stack that will not train

20 minutes

The specification

> **WHAT WE ARE ASKING FOR**
>
> A fully connected classifier: 3,072 inputs, **20 hidden layers** of 100 units, a logistic activation on each, and a linear head of 10 outputs. Cross-entropy loss, Adam at 0.001, batches of 128, 20 epochs.

Read that again and note what it does **not** say: nothing about how the weights start out. That is not something we are hiding — it is the ordinary case. Almost no specification mentions it.

The stack

```python
def make_net(depth=20, width=100, n_in=3072, n_out=10):
    layers, prev = [], n_in
    for _ in range(depth):
        layers += [nn.Linear(prev, width), nn.Sigmoid()]
        prev = width
    layers.append(nn.Linear(prev, n_out))
    return nn.Sequential(*layers)

net = make_net()
```

Nine lines. It is the same `nn.Sequential` from the previous lecture with the middle repeated 20 times.

Count the parameters before you run it

Layer	Shape	Parameters
First	3,072 → 100	307,300
Each of the other 19	100 → 100	10,100
Head	100 → 10	1,010
✓ Total	**✓ 21 weight matrices**	**500,210 ✓**

Comparable to the previous lecture's network, which reached 88% on Fashion MNIST. Nothing here is under-parameterised.

Why 20 layers

Because more layers is supposed to be more expressive. Each layer composes a new nonlinearity on top of the last, and the number of regions a network can carve out of its input space grows with depth faster than with width.

That is the theory. Today we test it.

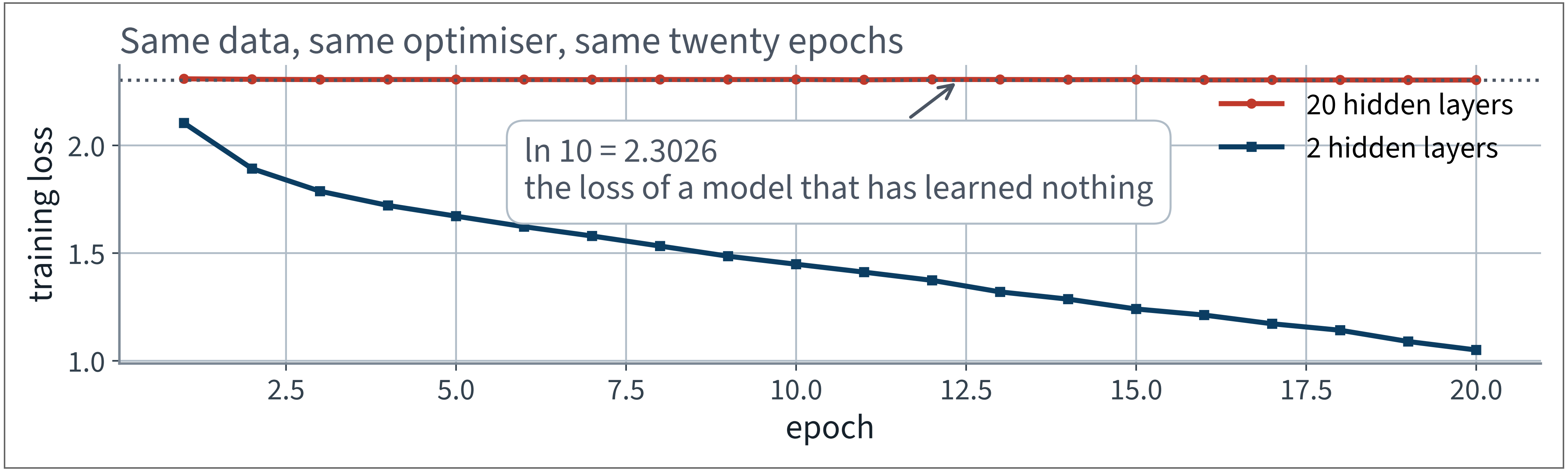
Nobody deploys a twenty-layer dense network on flattened pixels. It is here because it is the smallest thing that breaks in the way Chapter 11 is about, and because the break is invisible at three layers.

Run it

	Training loss
A model that has learned nothing, $\ln 10$	2.3026
Epoch 1	2.3091
Epoch 20	**✗ 2.3036**

8 seconds on mps, 20 epochs, 500,210 parameters.
The loss moved by 0.0055 in 20 epochs.

Same data, same optimiser, same twenty epochs



✗ The red curve is not noisy, or slow, or plateauing early. It has not started.

The number

10.0%

Test accuracy, 10,000 images.

Always predict one class	10.0%
✗ 20 hidden layers, 500,210 parameters	**✗ 10.0%**

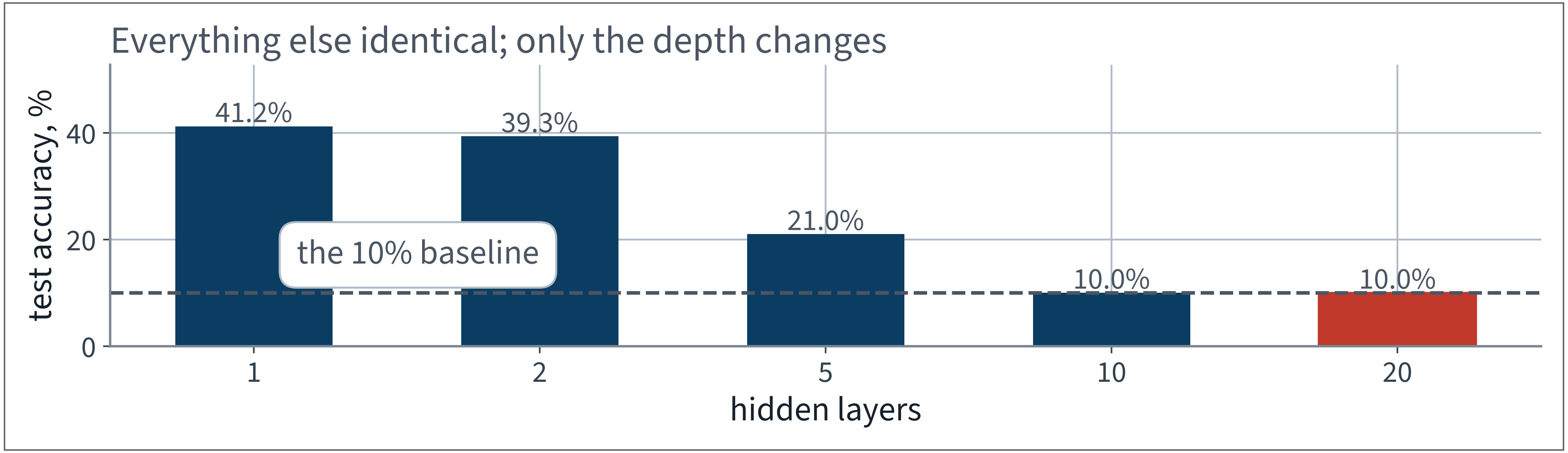
Half a million parameters and twenty epochs bought exactly what a single `return 3` would have.

Change one thing: the depth

```python
for k in (1, 2, 5, 10, 20):
    net, hist = train(make_net(depth=k))     # same seed, data, epochs
    print(k, accuracy(net, X_test, y_test))
```

Same data, same seed, same optimiser, same 20 epochs. One argument moves.

Accuracy against depth



✗ Adding layers made it worse, and past a point made it useless. That is not what capacity is supposed to do.

The sweep, as numbers

Hidden layers	Parameters	Final loss	Test accuracy
1	308,310	1.0259	**41.2% ✓**
2	318,410	1.0503	**39.3% ✓**
5	348,710	1.6925	21.0%
10	399,210	2.3035	**✗ 10.0%**
✗ 20	500,210	**✗ 2.3036**	**✗ 10.0%**

Note that the parameter count barely moves after the first layer — the 3,072-input layer dominates. So this is not a capacity effect.

What we know so far

- The loop is correct — it memorises 200 images on demand
- The data is correct — a shallower network on the same data reaches 39.3%
- The metric is correct — counted over the set, in evaluation mode
- **✗ The depth is the variable, and we have no account of why**

Which is where the previous lecture's real purchase pays off.

PART THREE, CONTINUED

Instrument it

the reason you own the loop

What we can do now, and could not before

Backpropagation computes 500,210 partial derivatives on every batch. Under `MLPClassifier.fit()` every one of them was discarded.

They are all still here, in `p.grad`, and we can look at any of them.

- **Forward** — what does each layer output?
- **Backward** — what does each weight matrix receive?

Two probes, about ten lines each. Without them the diagnosis today is “it does not train” and there is no next step.

Probe 1 · the forward pass

```python
@torch.no_grad()
def activation_stats(net, X, n=512):
    net.eval()
    h, rows = X[:n], []
    for m in net:
        h = m(h)
        if isinstance(m, nn.Sigmoid):
            rows.append({"mean": float(h.mean()),
                         "sd":   float(h.std()),
                         "sat":  float(((h - .5).abs() > .45)
                                       .float().mean())})
    return rows
```

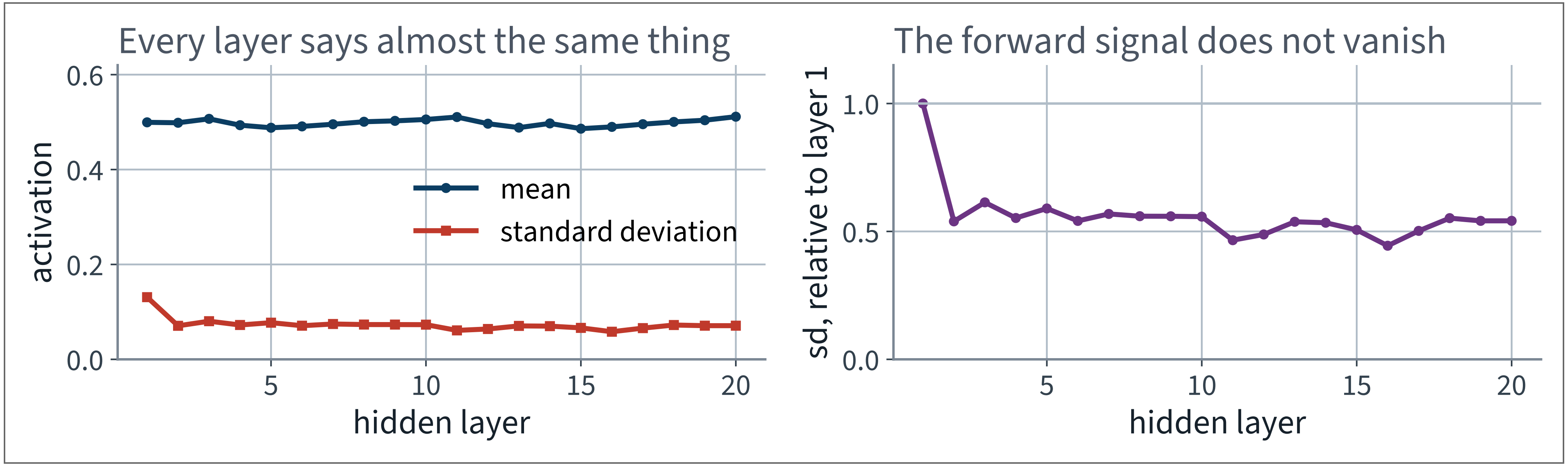
Running the modules by hand instead of calling `net(X)`. That is legal because a `Sequential` is an iterable of modules, and it is the cheapest possible instrumentation.

What every layer outputs

Layer	Mean	Standard deviation	Saturated
1	0.4995	0.1311	0.000
2	0.4986	0.0707	0.000
5	0.4881	0.0773	0.000
10	0.5055	0.0731	0.000
15	0.4861	0.0663	0.000
20	0.5114	0.0710	0.000

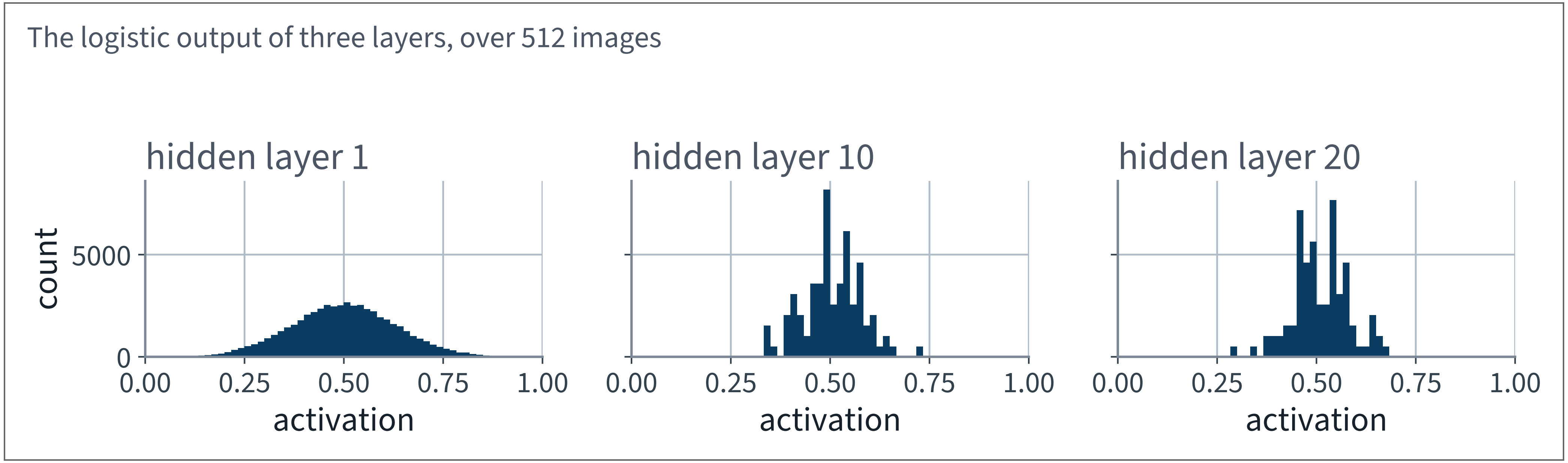
“Saturated” means the logistic’s own derivative has collapsed — here, an output further than 0.45 from a half.

The probe says the forward signal is fine



Mean near a half, spread near 0.071, flat for twenty layers. **Hold that conclusion for two slides.**

And nothing is stuck at the ends



If the logistic were saturating, these would be two spikes at 0 and 1. They are not. The textbook's first explanation does not apply to this network.

The probe measured the wrong thing

`h.std()` spans the *whole tensor*, so it is dominated by the spread of the layer's random **biases across units** — which does not change with depth. It would read 0.07 in a network that had stopped computing entirely. What carries information is the spread **down the batch** — how much a unit moves when the *input* does.

Hidden layer	1	5	10	20
`h.std()` — what we plotted	0.131	0.071	0.071	0.071
✗ **`h.std(dim=0).mean()` — the signal**	**1.3e−01**	**5.9e−05**	**3.0e−09**	**2.4e−17**

✗ Sixteen orders of magnitude. The forward signal does die **— by a factor of** 0.149 **per layer, which is the number the derivation predicts from the fan-in and the same rate the gradient vanishes at. By layer 20 the network is no longer a function of its input.**

✗ And the saturation column was no better. It asks whether $|h - \frac{1}{2}| > 0.45$, in a band whose sd is 0.071 **— a threshold** 6.3 standard deviations **away, and 3.4 even at layer 1.** ***Nothing could ever have exceeded it.*** **"0.000 at every depth" was arithmetic, not evidence.**

Probe 2 · the backward pass

```python
lins = [m for m in net if isinstance(m, nn.Linear)]

for _ in range(8):                             # 8 batches, not 1
    net.zero_grad()
    lossf(net(xb), yb).backward()
    acc += np.array([float(m.weight.grad.norm()) for m in lins])

profile = acc / 8
```

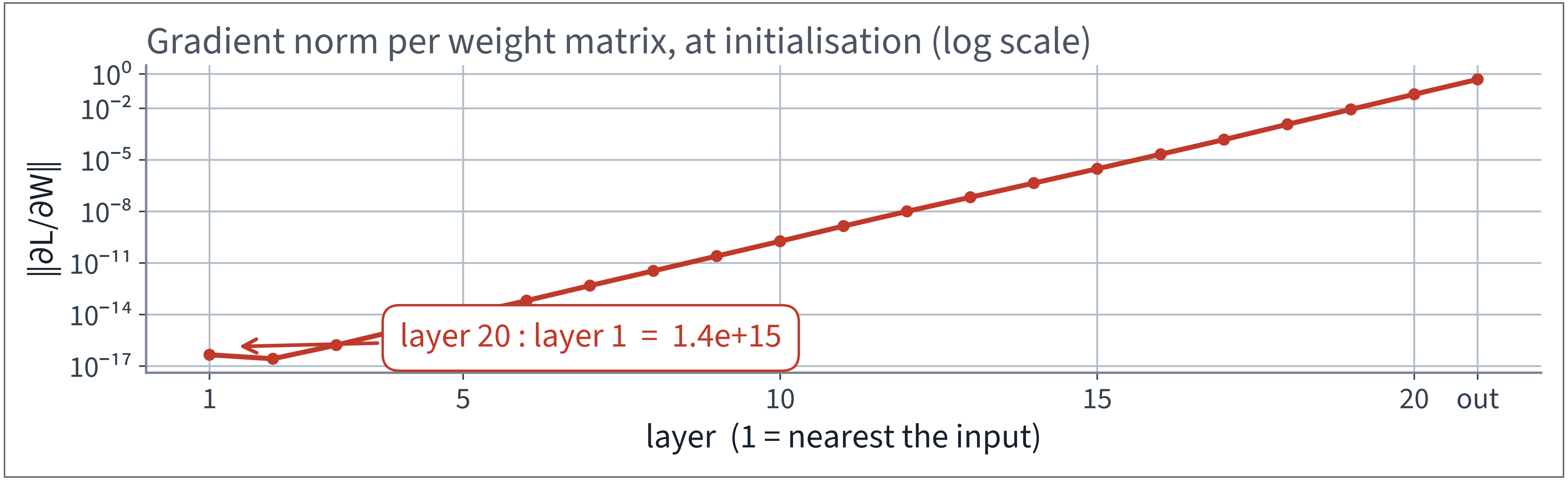
Eight batches rather than one, because a single batch of 128 is a noisy estimate of anything and a course that says so should not then quote one.

What each weight matrix receives

Layer	‖∂L/∂W‖
1 — nearest the input	**✗ 4.567e-17**
5	**✗ 9.296e-15**
10	**✗ 1.845e-10**
15	3.042e-06
20	6.608e-02
✓ head	**4.866e-01 ✓**

✗ Layer 20 receives 1.45e+15 times what layer 1 does.

Gradient norm per weight matrix, at initialisation



✗ Read the axis. Each gridline is a factor of ten, and the line crosses about 15 of them.

The shape of that line is the finding

It is **straight** on a logarithmic axis.
A straight line on a log axis is a geometric sequence. The attenuation is not one bad layer, or a fault at the input: it is the *same factor, applied at every layer*.

$$\frac{||g_\ell||}{||g_{\ell+1}||} \approx \rho \qquad \Longrightarrow \qquad \frac{||g_1||}{||g_{20}||} \approx \rho^{19}$$

Which means one number characterises the whole network, and it is a number the next lecture predicts from the shapes of the matrices.

The one number to take away

0.1593

The geometric mean of the ratio between neighbouring layers, over 19 steps.

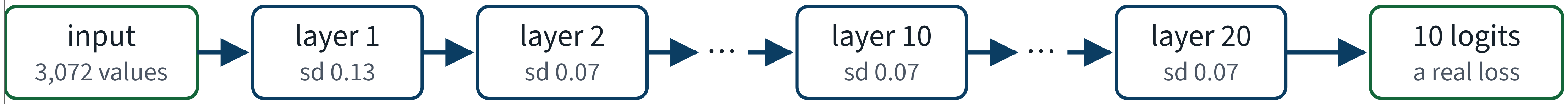
Per layer, descending	× 0.1593
Over 19 layers	**✗ × 6.91e-16**
Which is one part in	**✗ 1.45e+15**

Write both numbers on your sheet. The next lecture derives the first from arithmetic on the shapes of the weight matrices, and the second follows.

Both directions, on one picture

Forward: the signal survives

Each layer’s output looks like the one before it — same mean, same spread



Backward: the gradient does not

The same twenty layers, travelling the other way. Read this row right to left.

Every arrow multiplies the gradient by about 0.159.
Nineteen arrows later, layer 1 receives one part in 1.4×10^{15}.

Same twenty layers, same weights. Healthy going right, extinguished going left.

Does training rescue it?

A fair question: perhaps the first layers are merely slow to start, and the profile flattens as the network learns. We logged the per-layer norms at every epoch, so this is a lookup and not a guess.

Gradient at layer 1	‖∂L/∂W‖
Epoch 1	**✗ 6.660e-17**
Epoch 20	**✗ 4.610e-17**

20 epochs of Adam did not move the first layer’s gradient into a range where a learning rate of 0.001 could do anything with it.

What the weights actually did

Layer	Relative change over 20 epochs
✗ 1	**✗ 0.0000**
✗ 10	**✗ 0.0001**
14	0.1592
✓ 20	**0.5890 ✓**
✓ head	**0.0954 ✓**

Measured as $||W_{\text{after}} - W_{\text{before}}|| / ||W_{\text{before}}||$. The bottom half did not move at all — layer 1 to four decimal places, layer 10 to three. Only the top six layers learned anything, and they had nothing useful beneath them to work with.

Why Adam does not rescue this

Adam divides by a running estimate of the gradient's own magnitude, so it is tempting to think a small gradient is automatically compensated.

- It rescales, so the *direction* is what matters — and at 10^{-17} the direction is dominated by batch-to-batch noise
- The ε in the denominator, default 10^{-8}, is larger than the gradients themselves at the bottom of the stack

> An optimiser can compensate for a badly scaled gradient. It cannot manufacture information that the backward pass did not deliver.

“Just raise the learning rate”

The obvious reply, and it is worth stating why it cannot work.
The gradient at the head is 4.87e-01 and at layer 1 it is 4.57e-17. Any learning rate large enough to move layer 1 is 1e+15 times too large for the head.

ONE KNOB, TWO DEMANDS

A single scalar cannot serve a quantity that spans 15 orders of magnitude across the network. The repair has to change the *spread*, not the scale.

The chapter names two failures, not one

	Vanishing	Exploding
Per-layer factor	$\rho < 1$	$\rho > 1$
Symptom	the loss does not move	`NaN`, or a loss that jumps
Which layers suffer	the ones nearest the data	all of them, at once
How obvious	**✗ silent**	**✓ loud**

We have the quiet one.
Both are the same mechanism at different values of one number, which is why one derivation in the next lecture covers both.

The diagnosis, as far as we can take it

WHAT WE MEASURED

The gradient is attenuated by a constant factor of about 0.159 at every layer, so by the time it reaches the bottom of a 20-layer stack it has been multiplied by 1.4e+15. The layers nearest the data never learn anything.

What we cannot yet say: **why that factor is 0.159 and not 1.**

That is a question about the weight matrices themselves, and it has an exact answer.

THE MATHEMATICS

Variance propagation through layers

20 minutes · examinable

The question

Why 0.1593?

Not “why is it small”. Why is it *that number*, and what would it have to be instead?

> WHAT WE ARE GOING TO DO
>
> Track one quantity — the **variance** of what each layer passes on — forwards and then backwards. Two lines of algebra each way, and the answer falls out with the number in it.

One layer, written out

Drop the activation for a moment and look at the linear part alone:

$$z_i = \sum_{j=1}^{n_{\text{in}}} w_{ij}\, a_j$$

a is what arrives, z is what leaves, n_{in} is the fan-in — the number of inputs each output unit sums over. For our hidden layers that is 100; for the first layer it is 3,072.

Three assumptions, stated so they can be checked

1. The weights are drawn **independently**, with mean zero and variance $\mathrm{Var}(w)$
2. The weights are **independent of the inputs** — true at initialisation, and only approximately true afterwards
3. The inputs have mean zero and are uncorrelated with each other

All three are idealisations. That is fine: the point of an idealisation is that you can compute with it, and then check the answer against a measurement. We will do exactly that in twelve slides.

Assumption 3 is why we standardised the pixels in the previous lecture rather than dividing by 255 and leaving them in $[0, 1]$.

The one recursion

Under those assumptions each term $w_{ij}a_j$ has mean zero, and the terms are uncorrelated, so the variances add:

$$\mathrm{Var}(z) \;=\; n_{\text{in}} \cdot \mathrm{Var}(w) \cdot \mathrm{Var}(a)$$

THE FORWARD VARIANCE RECURSION

Everything in this lecture is a consequence of that line.

Note the shape of it: the input variance is *multiplied* by a factor. That is what will make depth dangerous.

Check it before believing it

```
Var(w) = 0.001   predicted Var(z) = 0.1000   measured 0.1002
Var(w) = 0.010   predicted Var(z) = 1.0000   measured 0.9727
Var(w) = 0.020   predicted Var(z) = 2.0000   measured 2.0502
Var(w) = 0.050   predicted Var(z) = 5.0000   measured 5.0050
```

One layer, fan-in 100, twenty thousand random inputs of unit variance. Four lines of NumPy, and the same habit as Lecture 2's orthogonality test: *verify the identity you are about to reason with*.

What the forward pass wants

We want the signal to arrive at layer 20 with the spread it had at layer 1. Otherwise, whichever way it drifts, twenty layers of drift is a factor of r^{20}.

$$\begin{aligned} \mathrm{Var}(z) = \mathrm{Var}(a) \quad &\Longleftrightarrow \quad n_{\mathrm{in}}\,\mathrm{Var}(w) = 1 \\ &\Longleftrightarrow \quad \mathrm{Var}(w) = \frac{1}{n_{\mathrm{in}}} \end{aligned}$$

One equation, one unknown. If the fan-in were the only thing in the network, we would be finished.

What the backward pass wants

The backward pass runs the same matrix **transposed**. Using the adjoint notation from thread 6:

$$\bar{a} \;=\; W^{\mathsf{T}}\bar{z} \qquad\Longrightarrow\qquad \mathrm{Var}(\bar{a}) \;=\; n_{\mathrm{out}} \cdot \mathrm{Var}(w) \cdot \mathrm{Var}(\bar{z})$$

Identical algebra with **one index swapped**: the transpose sums over the output dimension, so the fan-*out* appears where the fan-in did.

$$\mathrm{Var}(w) = \frac{1}{n_{\mathrm{out}}}$$

WHAT PRESERVING THE GRADIENT ASKS FOR

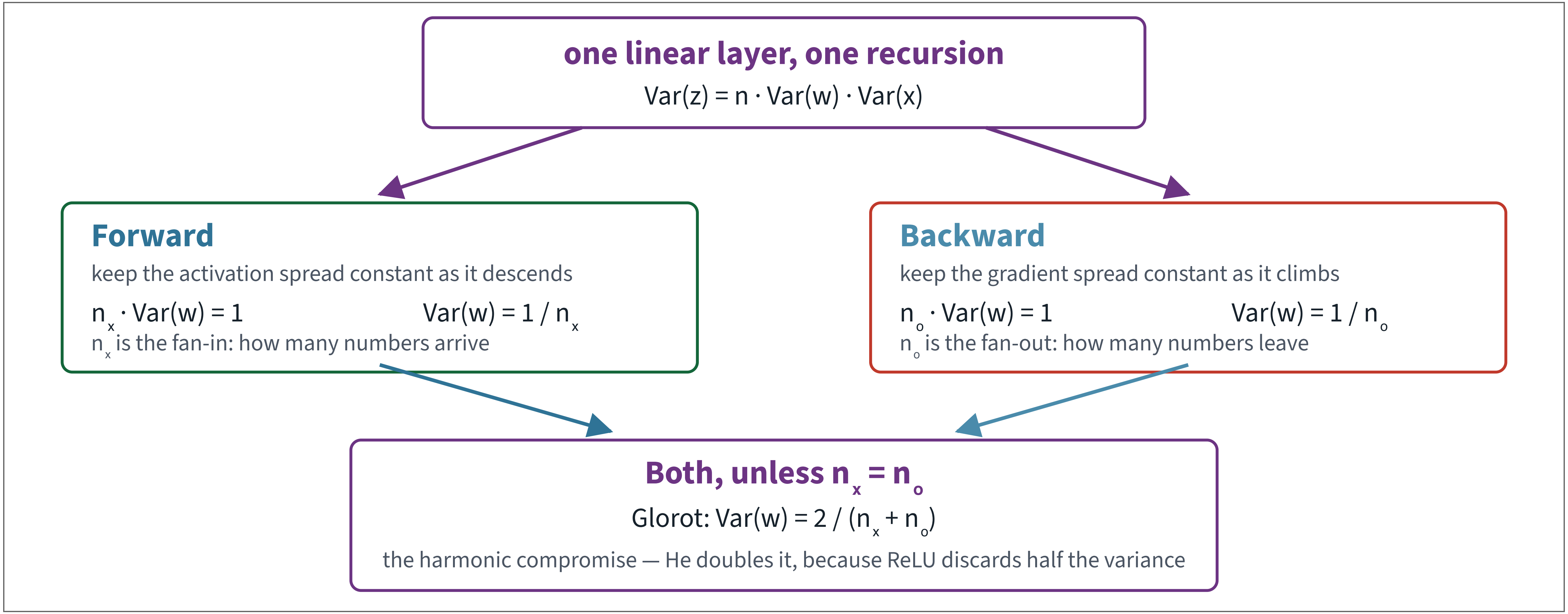
Why the index swaps

Because a row of W and a column of W are different sums.

- Forward, each output unit sums over n_{in} terms — one per incoming connection
- Backward, each input unit collects from n_{out} terms — one per outgoing connection

Exactly the fan-in / fan-out bookkeeping from thread 6: a node's adjoint is the sum over its outgoing edges. Here we are counting how many such edges there are.

Two requirements from one recursion



The same line of algebra, read in two directions, asks for two different answers.

The conflict

$\dfrac{1}{n_{\text{in}}}$ and $\dfrac{1}{n_{\text{out}}}$ are the same number only when $n_{\text{in}} = n_{\text{out}}$.

Layer	n_{in}	n_{out}	Can both be satisfied?
First	3,072	100	**✗ no — they differ by a factor of 30**
Each hidden	100	100	**✓ yes, exactly**
Head	100	10	**✗ no**

Every network whose layers change width — which is every network — has to give something up.

Glorot’s compromise

Do not choose. Take the **harmonic mean** of the two demands:

$$\mathrm{Var}(w) = \frac{2}{n_{\mathrm{in}} + n_{\mathrm{out}}}$$

GLOROT (XAVIER) INITIALISATION

A small error in both directions rather than a large error in one. On a layer where the two fans are equal it is not a compromise at all — it is exact, and reduces to $1/n$.

Harmonic rather than arithmetic because the quantities being averaged are reciprocals of the fans.

Two forms of the same thing

```
nn.init.xavier_normal_(w)      # N(0, 2/(fan_in + fan_out))
nn.init.xavier_uniform_(w)     # U(-r, r),  r = sqrt(6/(fan_in + fan_out))
```

The uniform version has the same variance: a uniform on $(-r, r)$ has variance $r^2/3$, and $6/3 = 2$. Choose either; nothing in the derivation cares about the shape of the distribution, only its variance.

On our hidden layers this gives $\mathrm{Var}(w) =$ 0.0100, a standard deviation of 0.1000 — against the default's 0.0577.

Now put the activation back

Between two linear maps sits φ, and the backward pass multiplies by its derivative pointwise. So the true per-layer factor on the *standard deviation* of the backward signal is

$$\rho = \sqrt{n_{\text{out}} \cdot \mathrm{Var}(w) \cdot \mathbb{E}[\varphi'(z)^2]}$$

THE NUMBER WE MEASURED AN HOUR AGO

Two factors we control — the shape of the matrix and the variance we initialise it with — and one that belongs to the activation function.

The logistic's contribution

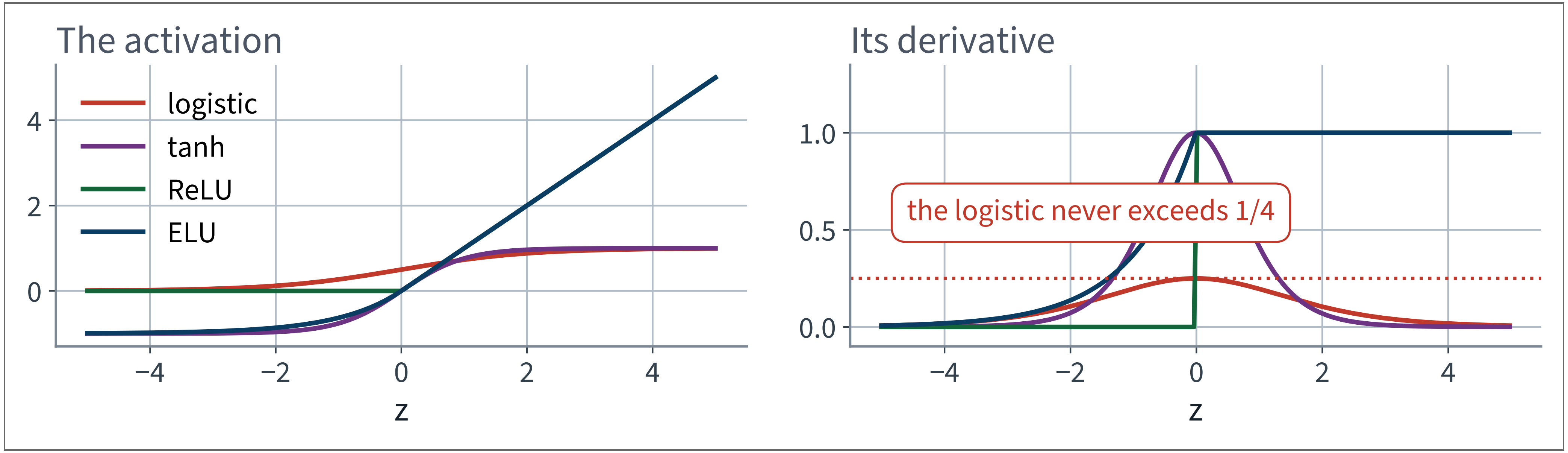
$$\sigma'(z) = \sigma(z)\big(1 - \sigma(z)\big) \;\le\; \tfrac{1}{4} \quad \text{for every } z$$

Not “usually small”. **Bounded above by a quarter, everywhere**, and equal to a quarter only at $z = 0$.

✗ So the logistic multiplies ρ by at most 0.25, before any weight has been chosen.

That is the ceiling. Over 19 layers, even a perfectly initialised logistic stack cannot do better than $(1/4)^{19}$.

Four activations and their derivatives



Read the right-hand panel. Everything except the logistic reaches a derivative of 1 somewhere; the logistic never does.

ReLU discards exactly half the variance

For a symmetric, zero-mean z, ReLU zeroes half the units and leaves the rest alone:

$$\mathbb{E}\left[\mathrm{ReLU}(z)^2\right] \;=\; \tfrac{1}{2}\,\mathbb{E}\left[z^2\right], \qquad \mathbb{E}\left[\mathrm{ReLU}'(z)^2\right] \;=\; \tfrac{1}{2}$$

Both statements are the same statement: half the time the derivative is 1, half the time it is 0. Nothing is bounded away from 1 the way the logistic is — the loss is a clean factor of two, and a factor of two can be paid for.

He’s adjustment

Pay for it by doubling the weight variance:

$$\operatorname{Var}(w) = \frac{2}{n_{\text{in}}}$$

HE INITIALISATION, FOR RELU

On our hidden layers, $\operatorname{Var}(w) = 0.0200$ and $\operatorname{sd}(w) = 0.1414$ — against Glorot's 0.1000 and the default's 0.0577.

✓ Then $\rho = \sqrt{n \cdot (2/n) \cdot \frac{1}{2}} = 1$, exactly.

Four schemes, predicted from the shapes alone

Scheme	$\mathrm{Var}(w)$	$\mathbb{E}[\varphi'^2]$	predicted ρ
✗ default, logistic	0.00333	0.0625	**✗ 0.1443**
Glorot, logistic	0.01000	0.0625	0.2500
Glorot, ReLU	0.01000	0.5000	0.7071
✓ He, ReLU	0.02000	0.5000	**1.0000 ✓**

Nothing in that table came from running anything. It is the fan-in, the fan-out, and one expectation.

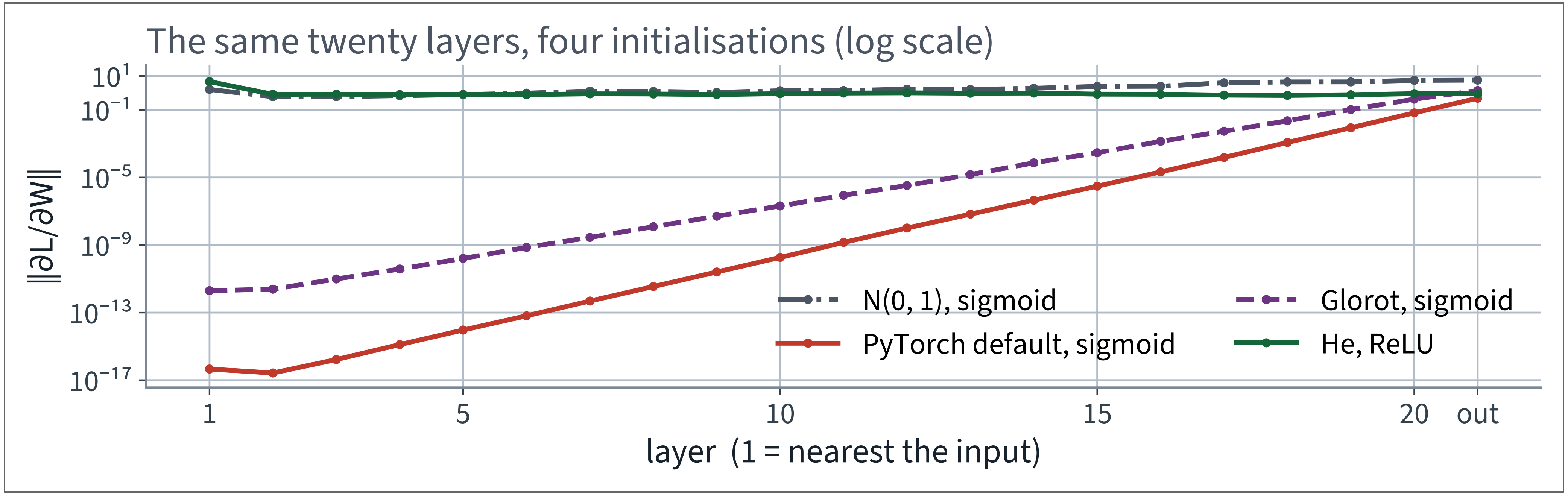
And then it compounds

$$\frac{||g_1||}{||g_L||} = \rho^{L-1}$$

There is no safe value of ρ except one.

ρ	over 19 layers	what you see	where
✗ 0.144	✗ 10^{-16}	**✗ the loss does not move**	**✗ the previous lecture**
0.707	10^{-3}	trains, slowly, badly	in forty minutes
1.000 ✓	**1 ✓**	**✓ every layer learns**	**✓ the target**
✗ 1.400	✗ 10^{3}	✗ `NaN`	illustrative

The same twenty layers, four initialisations



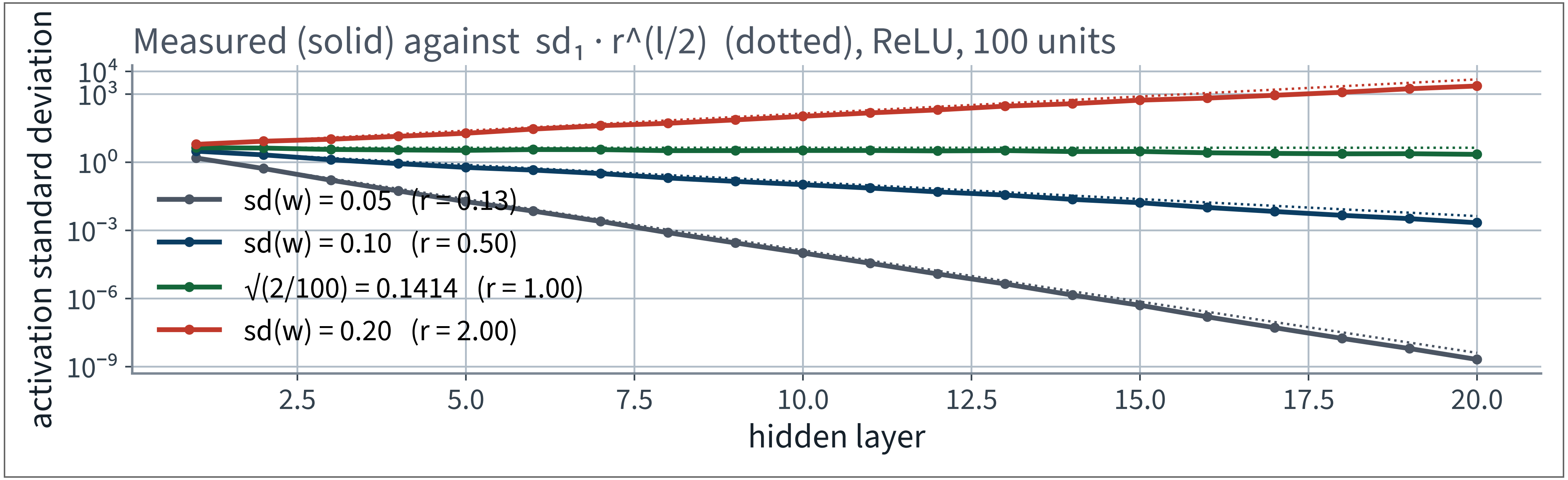
Four straight lines with four different slopes, and the slope is $\log \rho$ in each case.

The same argument, forwards

The recursion was symmetric, so the forward activations must be geometric in the depth too — with ratio $r = n_{\text{in}}\,\mathrm{Var}(w)/2$ for ReLU.

Easy to falsify: fix the activation to ReLU, set the weight standard deviation by hand, and watch the activation spread against depth. He's value for 100 units is $\sqrt{2/100} = 0.1414$.

Measured against $\mathrm{sd}_1 \cdot r^{\ell/2}$



The dotted lines are the prediction, not a fit. One value of the weight scale is flat; every other value is an exponential.

The derivation, summarised

1. One layer multiplies the variance by $n \cdot \mathrm{Var}(w)$ – forwards with the fan-in, backwards with the fan-out
2. Preserving the signal and preserving the gradient are the **same condition only when the fans are equal**
3. Glorot takes the harmonic mean of the two; He doubles it because ReLU discards half the variance
4. The per-layer factor compounds as ρ^L, so only $\rho = 1$ survives depth

Point 4 is the one that makes deep networks a separate subject rather than a bigger version of shallow ones.

THE METHOD

Seven repairs, measured one at a time

35 minutes · examinable

The notebook for this lecture

Open Lecture 11 in Colab

- **Runs on free CPU.** CIFAR-10 is subsampled and the stack is narrow; the point is the *shape* of the gradient curve, which survives the shrink
- The instrumentation: per-layer activation and gradient statistics, logged and plotted
- Each repair applied on its own, so the ablation is something you ran rather than something you read

> **WHAT TO CHANGE**
>
> Set the initialisation back to the default and re-run the gradient probe. The straight line on a log axis comes back, and its slope is the per-layer factor the derivation predicts.

Repair 1 · initialisation

```python
for m in net:
    if not isinstance(m, nn.Linear):
        continue
    nn.init.xavier_normal_(m.weight)     # Var = 2/(fan_in + fan_out)
    nn.init.zeros_(m.bias)

    # or, for ReLU:
    nn.init.kaiming_normal_(m.weight, nonlinearity="relu")
```

Two lines. `kaiming_normal_` is He; the argument `nonlinearity` is what selects the factor of two, and its default is `"leaky_relu"` with slope 0.01 — near enough, but say what you mean.

Glorot alone, measured

	ρ	End to end	Test accuracy
✗ Default, logistic	**✗ 0.1593**	**✗ 1.4e+15**	**✗ 10.0%**
Glorot, logistic	0.2531	2.2e+11	10.0%

✗ Eleven orders of magnitude better, and still hopeless.

Because the measured ρ is now 0.253 — which is 0.25, the logistic's own ceiling, with the weight scale perfectly neutralised. **You cannot initialise your way out of the activation function.**

Repair 2 · the activation function

Every requirement on φ comes from the thread:

- φ' should reach 1 somewhere, and not be bounded below it
- φ should not saturate on both sides, or gradients die where it does
- Output roughly centred on zero, so assumption 3 holds at the next layer

✓ **ReLU satisfies the first two and fails the third. It is still the right default.**

ReLU’s own failure mode

DYING UNITS

A unit whose pre-activation is negative for every input in the dataset has gradient exactly zero, forever. It never comes back.

The repairs are all the same idea — give the negative side a non-zero slope:

Variant	Negative side	Cost
Leaky ReLU	$\alpha z, \alpha \approx 0.01$	none
ELU	$\alpha(e^z - 1)$	an exponential per unit
SELU	ELU, scaled to self-normalise	strict conditions on the architecture

Six activations, measured

Activation	Initialisation	Test accuracy
logistic	Glorot	40.4%
tanh	Glorot	38.5%
ReLU	He	36.8%
Leaky ReLU	He	37.1%
✓ ELU	He	**43.2%** ✓
SELU	Glorot	41.0%

Spread: 6.4% points — and the logistic is not last.

All six **with batch normalisation**, which is why the logistic survives: normalisation puts the pre-activations back where its derivative is largest, at every step. Without normalisation, moving off the logistic was worth 31.1% points. With it, the choice of activation is worth 6.4%.

Repair 3 · batch normalisation

Initialisation gets ρ right once. Normalisation makes it right at every step, by standardising each layer's inputs and then learning what scale they should have had.

$$\hat{z} = \frac{z - \mu_B}{\sqrt{\sigma_B^2 + \varepsilon}}, \qquad y = \gamma \hat{z} + \beta$$

μ_B and σ_B are computed **over the batch**; γ and β are learned. The second step is what stops it from being a straitjacket: the network can undo the normalisation if it needs to.

Where it goes

```
layers += [nn.Linear(prev, width),
           nn.BatchNorm1d(width),        # before the activation
           nn.ReLU()]
```

- Before or after the activation is a genuine argument in the literature; the original paper puts it before, and so do we
- The bias in the `Linear` is now redundant — β does the same job. Harmless, and worth knowing

4,000 extra parameters across the whole network, against 500,210. Under one per cent.

The bill for batch normalisation

ITS STATISTICS DEPEND ON THE OTHER IMAGES IN THE BATCH

At training time the prediction for one image depends on which images happened to be batched with it. At evaluation time it uses running averages instead. Those are **two different functions**.

Which is Lecture 12's `model.eval()` failure again — and it is now much harder to catch. Dropout wobbles between calls, so running the evaluation twice exposes it. Batch normalisation in training mode is *deterministic* for a fixed batch, so the same trick finds nothing.

✗ New reviewer question: if there is a normalisation layer, was `eval()` called? Running it twice will not tell you.

Normalisation, measured

On the ReLU + He network	Test accuracy	Parameters	Seconds, 3 epochs
✓ No normalisation	**41.1% ✓**	500,210	5.24
✗ Batch normalisation	**✗ 36.8%**	504,210	**✗ 11.30**
✓ Layer normalisation	**41.6% ✓**	504,210	9.49

✗ Batch normalisation made it worse, by -4.3% accuracy points.

The timing is the *minimum* of 3 repeats: these figures were generated on a shared machine, contention can only make a run slower, so the smallest reading is the closest to the true cost. Batch normalisation costs 2.16× the epoch.

Why the repair everyone reaches for first did not pay

Two measurements, and they are not in conflict.

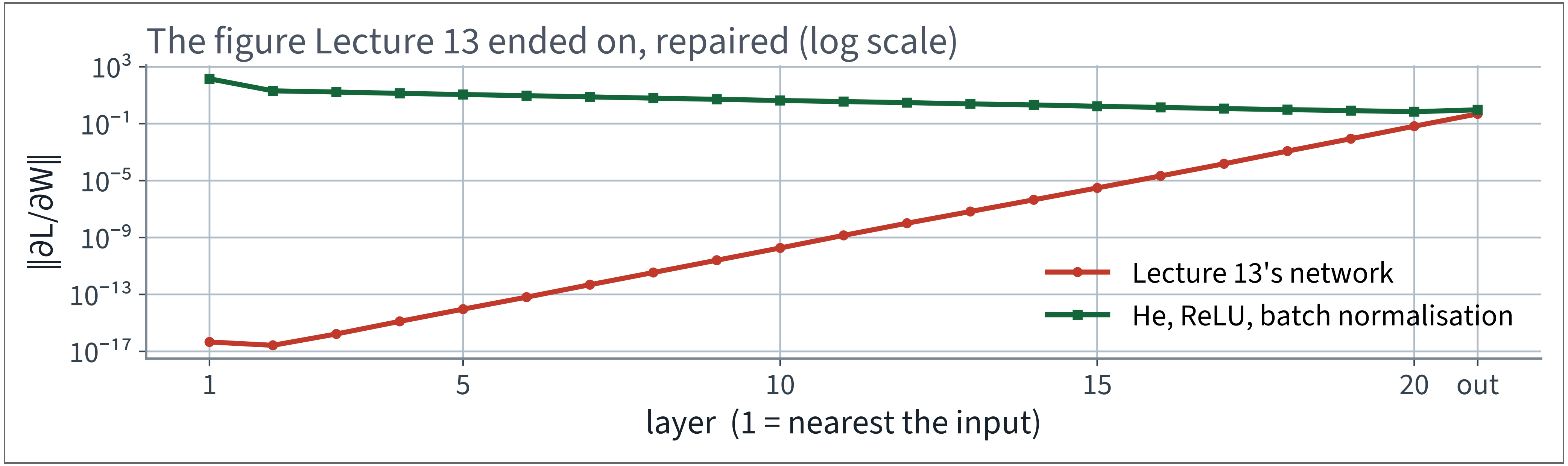
Batch normalisation applied to…	Test accuracy
✓ the *broken* network, on its own	**10.0% → 37.9% ✓**
✗ the *already repaired* network	**✗ 41.1% → 36.8%**

THE READING

Normalisation exists to hold ρ near 1 when something else has pushed it away. He initialisation had already put it there, so there was nothing left to correct — and what remained was the cost.

On a wider network, a longer run, or one whose weights drift further, that trade goes the other way. The point is that you can tell which case you are in, because you measured both.

The figure the previous lecture ended on, repaired



✓ Same axes, same scale, same network shape. Layer 1 now receives a gradient of the same order as layer 20.

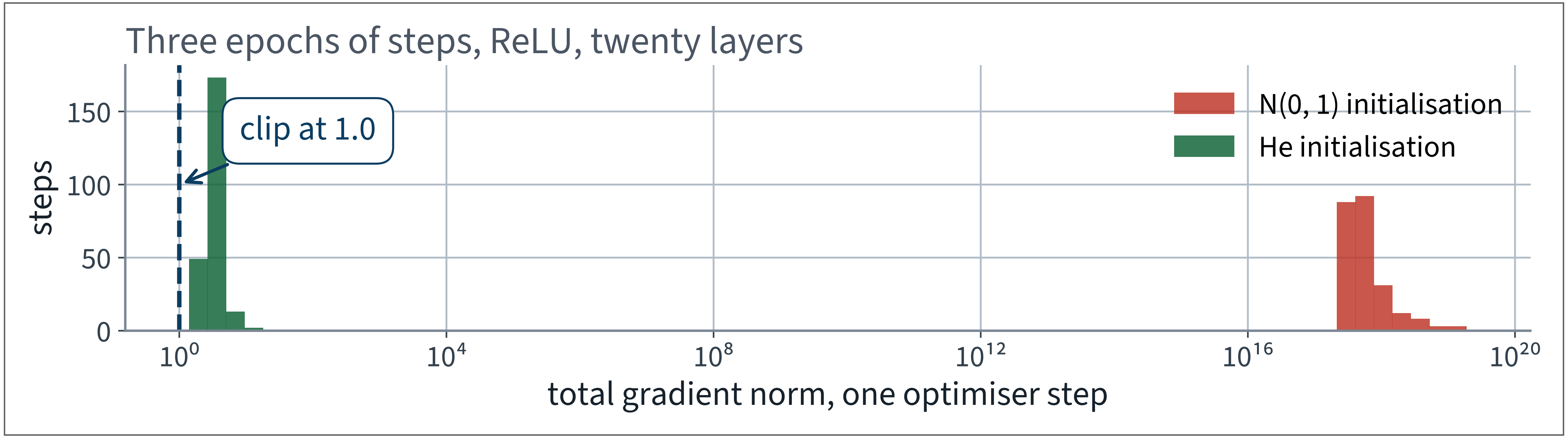
Repair 4 · gradient clipping

```
loss.backward()
torch.nn.utils.clip_grad_norm_(net.parameters(), 1.0)
opt.step()
```

Rescale the whole gradient vector when its norm exceeds a threshold. It preserves the direction and bounds the length.

✗ This is a defence against $\rho > 1$. We have $\rho < 1$. Look at the distribution of norms before choosing a threshold — a clip below the median silently turns your optimiser into something else.

What the step norms actually are



Median 3.19 with He, against a median of 4.6e+17 with N(0, 1). The threshold of 1.0 used in the ladder sits *below* the He median — deliberately, and the next slide says what that bought.

What clipping bought, and what it did not

Gradient clipping at 1.0	Test accuracy
✗ applied alone to the failing network	**✗ 10.0% — nothing**
✓ added to the repaired network	**36.8% → 39.6% ✓**

Alone it cannot help: you cannot bound your way out of a gradient that is 1e+15 times too *small*. On the repaired network a threshold below the median acts as a step-size limiter rather than a safety net, and here that was worth +2.8% points.

✗ Which is a useful effect obtained for the wrong reason. If you want a smaller step, set a smaller learning rate and say so.

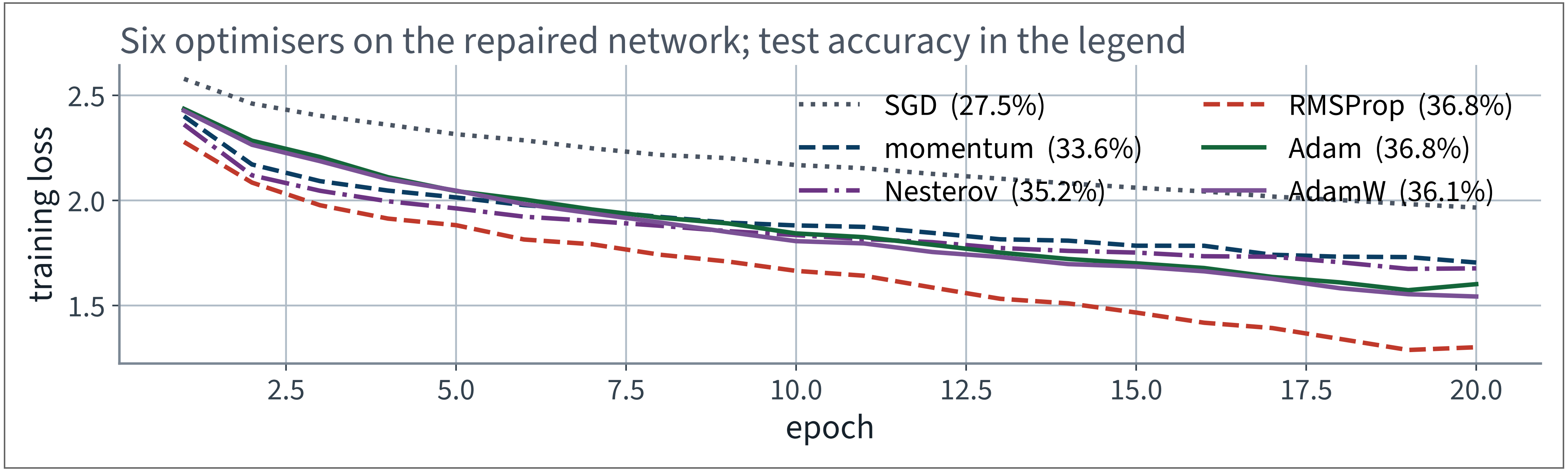
Repair 5 · faster optimisers

Now that the gradient reaches every layer, the optimiser has something to work with. Comparing them on the broken network would have measured nothing.

Optimiser	Idea	Test accuracy
SGD	step along the gradient	27.5%
Momentum	accumulate a velocity	33.6%
Nesterov	measure the gradient ahead of the step	35.2%
RMSProp	divide by a running gradient scale	36.8%
✓ Adam	momentum and RMSProp together	**36.8% ✓**
AdamW	Adam, with the weight decay decoupled	36.1%

SGD, momentum and Nesterov at 0.01; the adaptive three at 0.001 — the same learning rate would not be a fair comparison.

Six optimisers on the repaired network



The spread between the best and the worst optimiser here is far smaller than the effect of the initialisation. Order your effort by the measurement, not by the novelty.

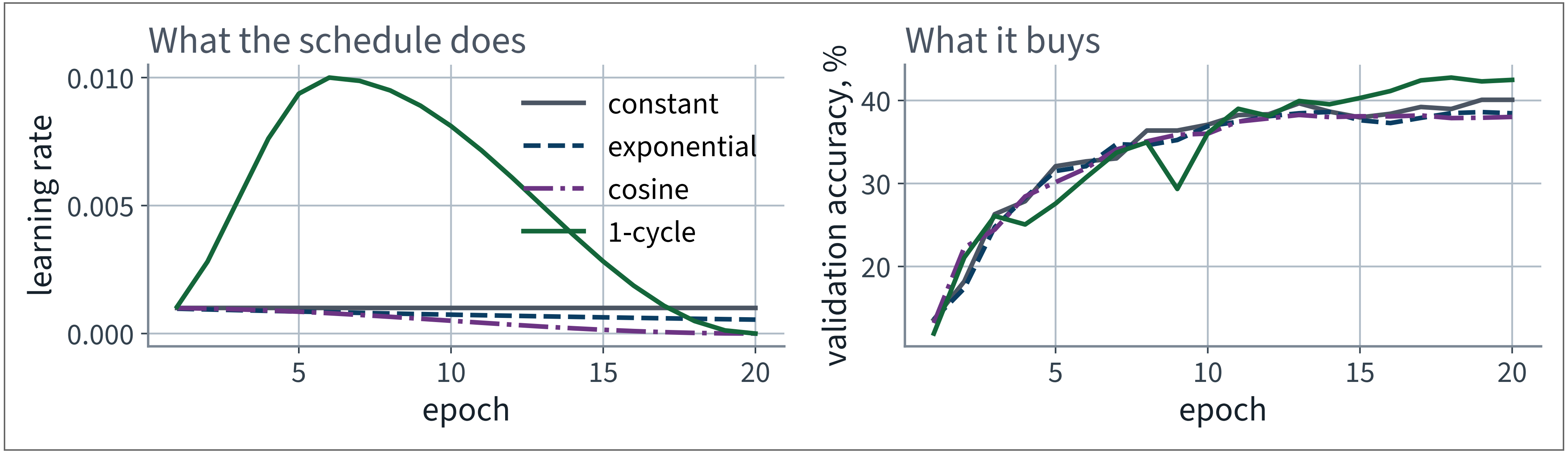
Repair 6 · learning-rate schedules

One step size cannot suit both the start of training and the end. The book gives five schedules; three are worth having.

- **Exponential** — multiply by a constant each epoch
- **Cosine** — anneal smoothly to zero over the run
- **1-cycle** — rise to a peak, then fall well below the starting value

All three need to know how long the run is, which is why they take `total_steps` and why changing the epoch count silently changes the schedule.

What a schedule does, and what it buys



Constant 39.6%, cosine 38.8%, 1-cycle 41.5% — on a twenty-epoch run, one seed each. In the ablation the same schedule is worth +0.5 points against a five-seed spread of ±2.79, so read this panel as a shape, not a ranking.

Repair 7 · dropout

Zero a random fraction of each layer's units during training, and scale the rest up so the expected value is unchanged.

```
layers += [nn.Linear(prev, width), nn.BatchNorm1d(width),
           nn.ReLU(), nn.Dropout(0.1)]
```

✗ This is a regulariser — it fights overfitting. A network sitting at 10% is not overfitting, so applying it to the failing network should do nothing. We are going to check that it does nothing rather than assume it.

THE LAST FIFTEEN MINUTES

Which repair actually paid

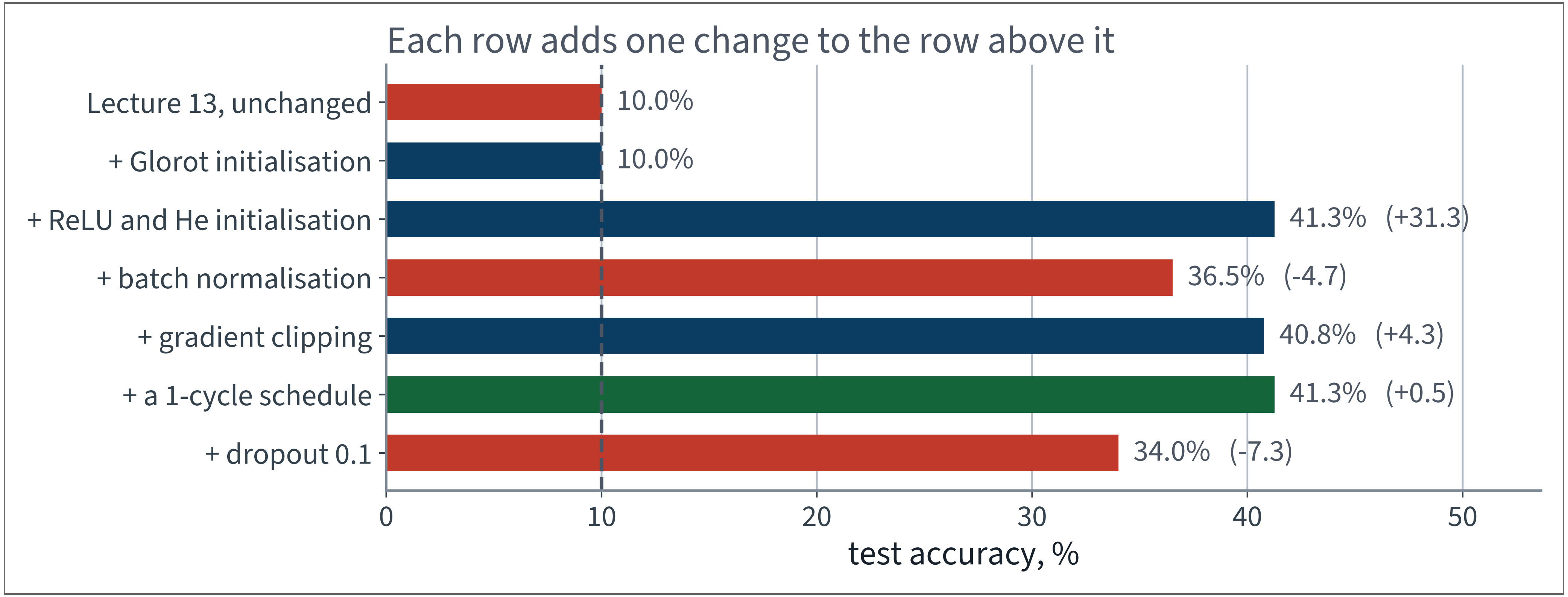
15 minutes

Each row adds one change to the row above

Configuration	Test accuracy	sd over 5 seeds	Change	Final training loss
✗ the failing network, unchanged	10.0%	±0.00	—	2.3035
+ Glorot initialisation	10.0%	±0.00	+0.0 (noise)	2.3062
✓ + ReLU and He initialisation	41.3%	±0.35	**+31.3 ✓**	1.1373
+ batch normalisation	36.5%	±0.64	**✗ -4.7**	1.5461
✓ + gradient clipping	40.8%	±0.76	**+4.3 ✓**	1.1064
+ a 1-cycle schedule	41.3%	±2.79	+0.5 (noise)	1.2773
+ dropout 0.1	34.0%	±1.33	**✗ -7.3**	1.6346

The deliverable of the lecture — photograph it, and photograph the third column with it. Five seeds a row; typical spread **0.84 points**. One step is worth **+31.3**, forty times that. Three more clear it. **The 1-cycle schedule does not** — +0.5 against ±2.79 — and the single-seed version of this table called it the winner.

The same table, as a picture



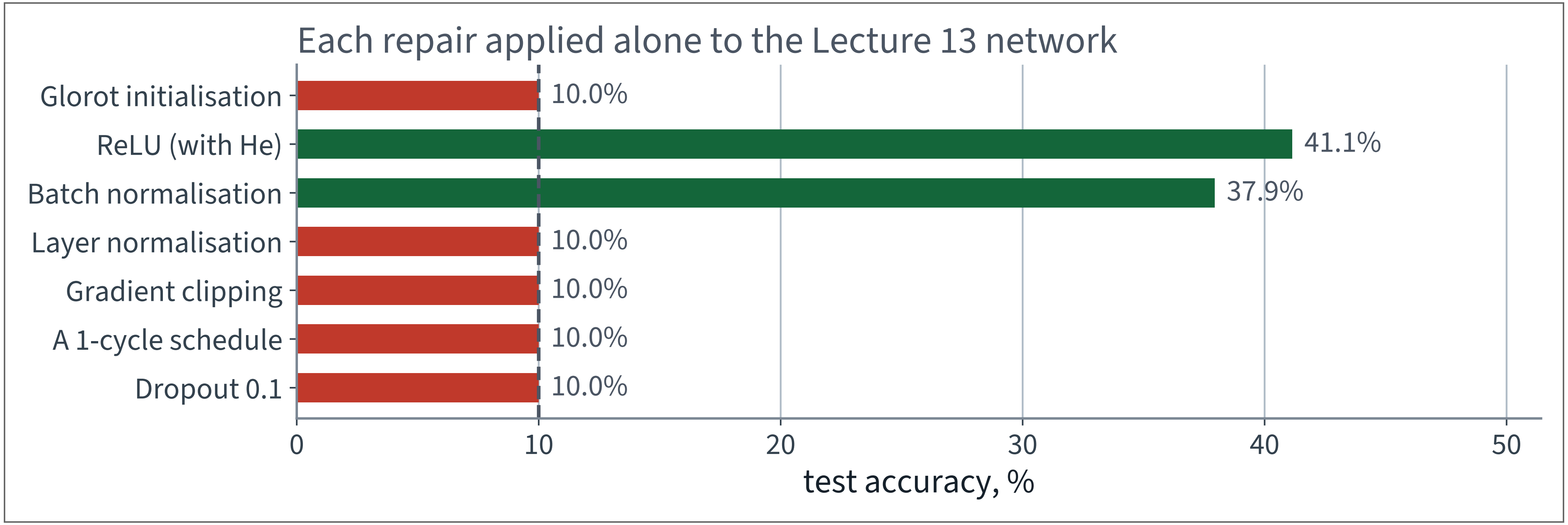
✓ One row moves the number further than the whole journey, and it is the row the mathematics pointed at.

Where the gain came from

	Accuracy points
✓ The largest single step — ReLU with He	**+31.3% ✓**
✓ Best configuration, against the start	**+31.3% ✓**
✗ The worst single step	**✗ -7.3%**
The last row, against the start	+24.0%

One change is worth more than the whole ladder.
Which is the answer to “should I add batch normalisation, or dropout, or a schedule?” — the question is malformed until you know which failure you have. Ours was $\rho \neq 1$, and only the repairs that address ρ paid.

Each repair alone, on the broken network



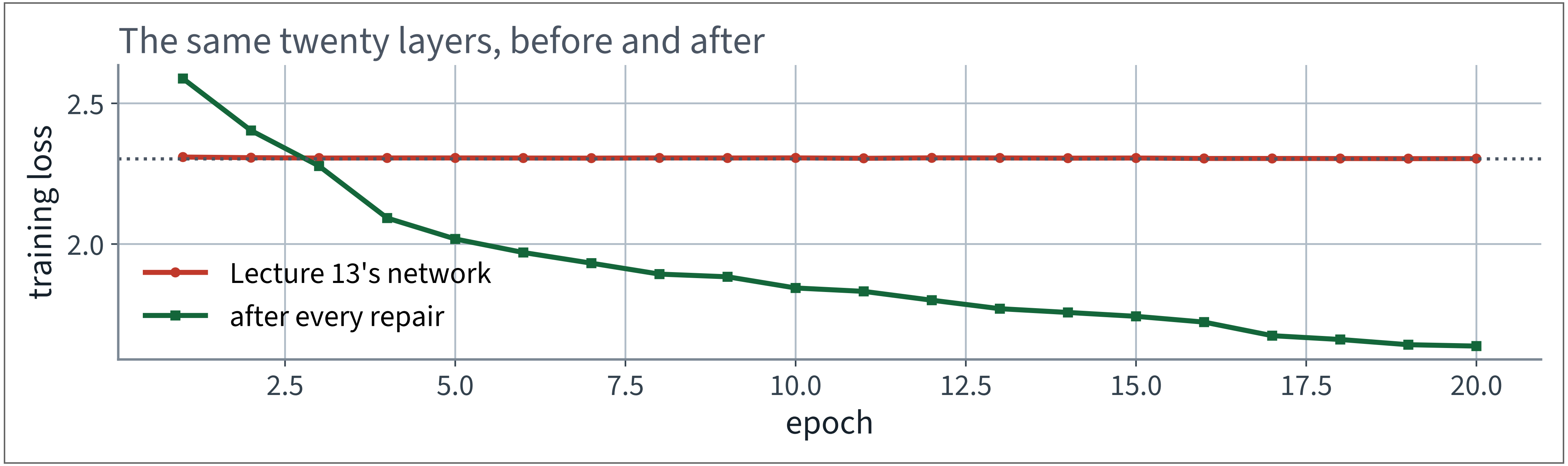
Four of these leave the network exactly where it was. That is not a disappointment — it is what tells you what each one is for.

One at a time, as numbers

Applied alone to the failing network	Test accuracy
Glorot initialisation	10.0%
✓ ReLU (with He)	**41.1% ✓**
✓ Batch normalisation	**37.9% ✓**
✗ Layer normalisation	**✗ 10.0%**
✗ Gradient clipping	**✗ 10.0%**
✗ A 1-cycle schedule	**✗ 10.0%**
✗ Dropout 0.1	**✗ 10.0%**

Clipping, the schedule and dropout answer problems this network does not have. Glorot buys eleven orders of magnitude of gradient and is still not enough.

The same twenty layers, before and after



✓ **Same architecture, same data, same optimiser, same epochs, same seed. Four lines changed.**

The silent-failure catalogue so far

Failure	Diagnosed in	Measured cost
Scaler fitted before the split	Lecture 2	about a dollar
Accuracy under class imbalance	Lecture 3	a useless model above 90%
Missing `optimizer.zero_grad()`	Lecture 10	76.7 points
Metric averaged per batch	Lecture 10	0.382 points
✗ A deep stack on default initialisation	**✗ today**	**✗ 31.3% points**
✗ Glorot’s constant used with ReLU	**✗ today**	**✗ 1.5% points**

Every one of them ran to completion and printed a plausible number. The last two are the first in which no line of code is wrong.

A checklist for a deep network

- Which initialisation, and derived for which activation?
- Is the per-layer gradient ratio within about 20% of 1?
- Do the fan-in and fan-out differ on any layer, and by how much?
- If there is normalisation: `train()` before training, `eval()` before every measurement?
- Was the clipping threshold chosen from the observed norms?
- Does the schedule know the true number of steps?
- Is every entry in the results table one change from its neighbour?

What we did

1. Instrumented a network that would not train, and found the gradients shrinking by a constant factor per layer — a straight line on a log axis
2. Derived why: a layer's output variance depends on fan-in and weight variance, and preserving the forward signal and the backward gradient impose *conflicting* requirements
3. Saw Glorot's compromise between them, and He's adjustment for ReLU, which halves the variance
4. Understood that the error compounds geometrically, so twenty layers multiply a per-layer factor twenty times
5. Applied seven repairs — initialisation, activation, normalisation, clipping, optimiser, schedule, regularisation — and measured each alone
6. Read the ablation, and found that most of the gain came from the first two

In the book

Topic	Chapter
Vanishing and exploding gradients	11
Glorot and He initialisation	11
Nonsaturating activation functions	11
Batch normalisation	11
Gradient clipping	11
Faster optimisers and learning-rate scheduling	11
Dropout and regularisation for deep networks	11
Layer normalisation	13

NOT PRESENTED — FOR AFTERWARDS

Exercises

Lecture 11

five, in the style of the written paper

Exercises — Lecture 11

1 One layer multiplies the standard deviation of the backward signal by $\rho = \sqrt{n_{\text{out}}\,\mathrm{Var}(w)\,\mathbb{E}[\varphi'^2]}$. State what L layers do to it, and why only $\rho = 1$ survives depth. [5]

2 Preserving the forward signal asks for $\mathrm{Var}(w) = 1/n_{\text{in}}$ and preserving the gradient asks for $1/n_{\text{out}}$. State when the two agree, and what Glorot does when they do not. [4]

3 He initialisation doubles the weight variance for ReLU. Derive the factor of two in one line. [4]

Solutions on Lecture 12’s deck.

Exercises — Lecture 11 (continued)

4 A gradient profile that is a straight line on a log axis tells you something a curved one does not. State what, and why it identifies the cause. [4]

5 Gradient clipping is applied to a network whose gradients are fifteen orders of magnitude too small. Predict the effect, and say what the attempt reveals about the diagnosis. [4]

Solutions on Lecture 12’s deck.

Solutions

Lecture 10

not part of this lecture

Solutions — Lecture 10

1. Reverse-mode automatic differentiation costs one sweep for all partial derivatives; forward mode costs one…

✓ Many inputs, one output — which is exactly a loss.

- Forward mode's cost scales with the number of inputs, reverse mode's with the number of outputs
- A neural network has millions of parameters and one scalar loss

2. Explain why the loss must be a scalar for .backward() to be called without arguments.

✓ The reverse sweep is seeded with $\partial L/\partial L = 1$, which only exists for a scalar.

- For a vector output there is no single derivative to seed with
- PyTorch then requires an explicit vector to weight the outputs by

Solutions — Lecture 10 (2)

3. The five lines of the training loop are reordered so that opt.step() comes before loss.backward(). State what…

✓ Nothing raises, and the model stays at chance.

- The step is taken with the gradients zeroed at the top of the iteration
- The gradients computed afterwards are cleared before they are ever used

4. A network with dropout is evaluated without calling model.eval(). Describe the symptom precisely, and the…

✓ The score wobbles between identical calls. Evaluate twice and assert the two agree.

- Dropout keeps sampling masks in training mode, so the metric is a random variable
- A deterministic function of fixed weights and fixed data does not change between calls

Solutions — Lecture 10 (3)

5. GPUs did not help at small model sizes in the measured comparison. Give the reason, and the quantity that…

✓ Transfer and launch overhead dominates; the crossover is set by arithmetic per byte moved.

- A small matrix product finishes faster than the cost of getting it there
- The device pays once the work per transferred byte is large enough

LECTURE 12 · GÉRON CH. 12

Convolutional networks

Weight sharing, equivariance, and where the memory goes — derived
Why a flattened image is the wrong representation
Convolution, pooling, and a network built from scratch

Run the Lecture 11 notebook first